8.1 UNDERGROUND MINING METHODS WITH STOWING IN COAL MINES OF INDIA

In India, underground coal mining methods with stowing are necessary because of various constraints usually faced in caving techniques, specifically, for extracting pillars in more than one section, extraction of thick seams, working of contiguous seams/sections and mining of steeply dipping coal seams.

As per Prasad and Rakesh (1992), while working any coal seam the following statutory provisions related to stowing are in force which need to be adhered:

a) Where splitting as a final operation along with stowing is adopted:

- The splits before stowing should have a minimum factor of safety 1.
- No splitting shall generally be permitted where the hard cover is less than 15 m.
- The minimum size of stooks shall be 6 m^2, where the depth is 135 m or more or the thickness of the seam is more than 7.5 m. In all other cases it should be kept at 4.5 m^2.

b) Where pillar extraction with stowing is done in more than one section, the bottom section shall be kept generally in advance (by not more than the half a pillar). In some cases, extraction may be simultaneous also.

c) When two seams/sections of a seam have been developed in close proximity:

- If the pillar in the either seams/sections are not too much out of coincidence vertically, the galleries in question may be widened so as to restore verticality. Then the pillars may be extracted simultaneously.
- If the verticality of the pillars and the galleries is too much out the pillars in the bottom seam/section may be split and the voids stowed. The voids in the top seam/section may then be caved in.

d) Where extraction of thick seams i.e. seams more than 4.8 m thick, with caving is not free from danger, because the quality of roof support available sharply deteriorates in long props. Such props have a diameter of only 80-100 mm at the top. They cannot afford adequate support to the roof, the extraction of seams more than 4.5 m in thickness, should be done only by hydraulic stowing.

8.1.1 Bord and Pillar Working with Stowing for Contiguous Seams/Sections

In Indian coal mining, contiguous seams refer to two or more coal seams that are situated close to each other, specifically within a parting of less than 9 m (CMR, 2017; Mandal et al., 2008; Singh et al., 2012). This close proximity means that mining one seam can significantly impact the stability and safety of the other. Hence, proper design of extraction and support system is required while extracting contiguous seams/sections. Safe underground extraction of contiguous seams/sections is an important issue for designers/planners, mainly due to their frequent occurrence and presence in most of the coalfields in India.

Multiple coal seams exist in many coalfields in India, for instance, the Jharia coalfield has the maximum number of coal horizons: as many as 40 (Saxena, 1991). In many cases, the coal seams become contiguous. As the Jharia coalfield is highly populated, the presence of several surface structures like buildings, road, pond, overhead tank, etc. make the extraction of these seams very difficult. Other countries of the world do not have many contiguous seams/sections.

Plenty of works related to extraction of contiguous seams/section with stowing have been done in India. For the extraction of the coal seams which are in contiguity, it is considered as safer if the extraction of the top seam is carried out first (considering caving). The bottom seam is extracted after the settlement of the goaf of the top seam (Mandal et al., 2008; Zhang et al., 2018a, 2018c). The coal seams extracted in this manner experience the transferring of abutment stresses from the overlying pillars, ribs and the barrier pillars. It is also very difficult to ensure the settlement of the goaf of the top seam, as there is no defined time and procedure to identify the complete settlement of the top seam goaf (Mandal et al., 2008). As the stress concentration is high at the remnant pillars kept in the goaved-out area of the top seam, it affects the safety of the parting during mining in the lower seam. There is a chance of accumulation of the gases and the percolation of water through the cracks in the partings during the extraction of the bottom seam. If the bottom seam is extracted first without stowing, the top seam experiences the cracks due to the effect of subsidence of the strata above the bottom seam.

According to Rakesh & Prasad (1992), while working a second or subsequent lift of a seam over a stowed area, pillars may be formed over the stowed goaf, and extracted immediately thereafter. However, while working contiguous seams with stowing, if the stowing is good and prompt, and lag is kept within limits, excessive trouble may not arise, and the extraction may be either kept simultaneous, or the bottom seam be kept in advance by not more than half a pillar. However, in case of a very strong parting, experimental permission may have to be obtained by the mine management for simultaneous extraction with stowing, keeping the top seam/section slightly in advance.

A typical example of extraction of contiguous seams with sand stowing is elaborated hereafter. Bhowra (North) colliery of Bharat Coking Coal Limited (BCCL), India is situated between 23° 41′ 07″ N to 23° 41′ 30″ N Latitude and86° 24′ 15″ to 86° 24′ 42″ E Longitude in the south-eastern part of the Jharia coalfields. The workings of contiguous V (bottom) and VI (top) seams at the mine are standing on pillars. The total thicknesses of the seam V and seam VI are 3.4 m and 4.1 m respectively (Fig. 8.1). The parting between the seams is 6 m. The size of the developed pillars in bottom seam V is 30 m × 30 m (center-to-center).

The width and height of the galleries of V seam are 4.2 m and 3.0 m respectively. The seam is developed along the floor, leaving 0.4–0.6 m coal in the roof. Similarly, the size of the developed pillars and width of the gallery in seam VI are same as seam V but the seam VI is developed with a 3.0 m height of the gallery leaving around 1.1 m coal in the roof. The depth of covers of the panel V ‘A’ in V seam and VI ‘A’ panel in VI seam are 127 m and 115 m respectively. There are several surface features like buildings, residential areas, and roads above the panels of both the seams. Since, the workings are below built-up areas, the method of pillar extraction in these two contiguous seams was such that the surface features above the workings remain safe after the extraction.

To extract the solid pillar of 30 m × 30 m in the bottom seam V, a split gallery of 4.5 m width is driven along the level in the middle of the pillar making the pillar into two equal halves. Then, a 4.5 m width dip split gallery is driven from the middle of the level split gallery making the pillar into four equal stooks, as shown in Fig. 8.2a. Full stowing is done in the level split gallery, dip split gallery and original gallery. In a similar manner, extraction is carried out in the other pillars maintaining a diagonal line of extraction.

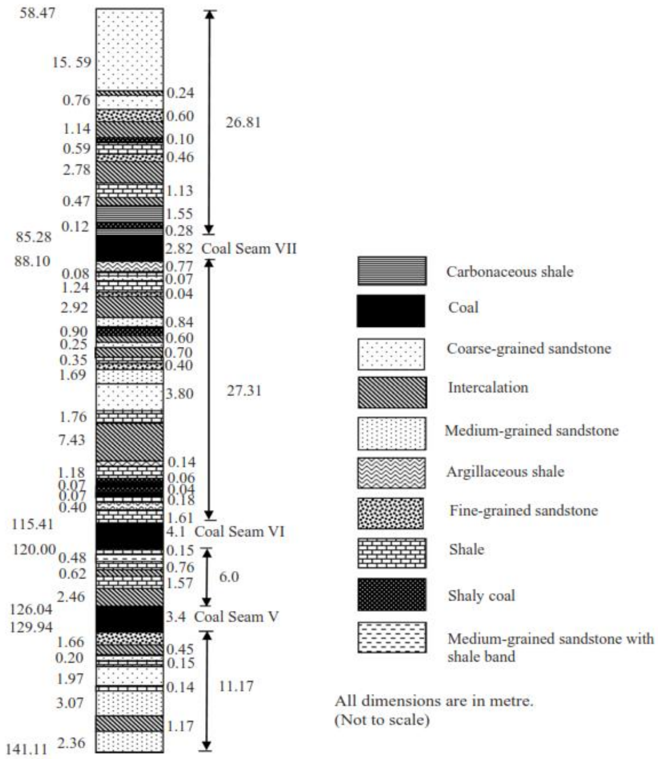
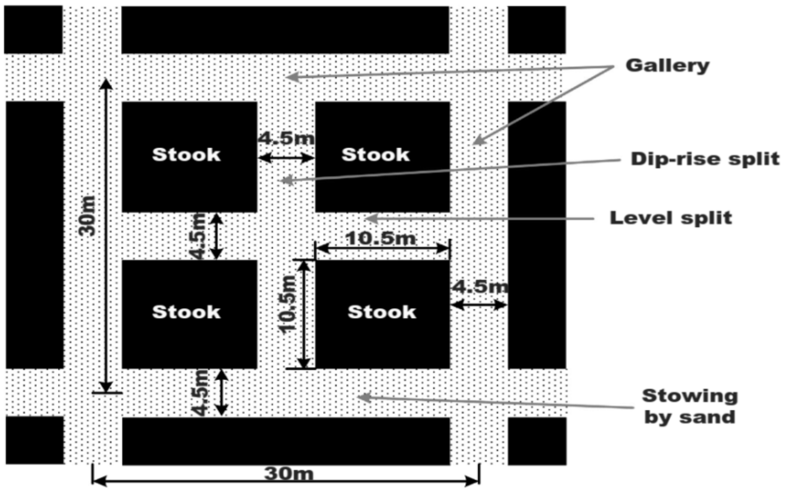
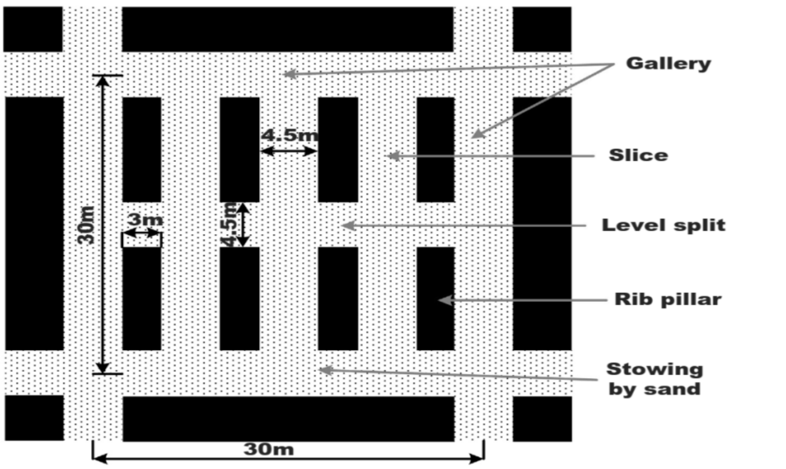


Fig. 8.1. Formation of different overlying and underlying strata of V and VI coal seams as per the borehole section NBW-19 (Das et al. 2020)

For top seam VI, the pillars are extracted with hydraulic stowing. The extraction operations are performed in two stages, in the first stage, the pillar of 30 m × 30 m size is divided into two halves by making a 4.5 m wide split gallery along the level in the middle of the pillar. The slices of 4.5 m width are taken while leaving ribs against the goaf, as shown in Fig. 8.2b. The rib pillar width is kept not less than 3.0 m. The diagonal sequence of extraction is followed for the purpose of pillar extraction in both the coal seams with full stowing. The working at the bottom seam V is kept one pillar advance of the working at the top seam VI as shown in Fig. 8.3. The voids of the bottom seam immediately below the workings at the top seam are stowed.



(a)



(b)

Fig. 8.2. Method of a coal pillar extraction for (a) bottom seam V and (b) top seam VI with hydraulic sand stowing (Das et al. 2020).

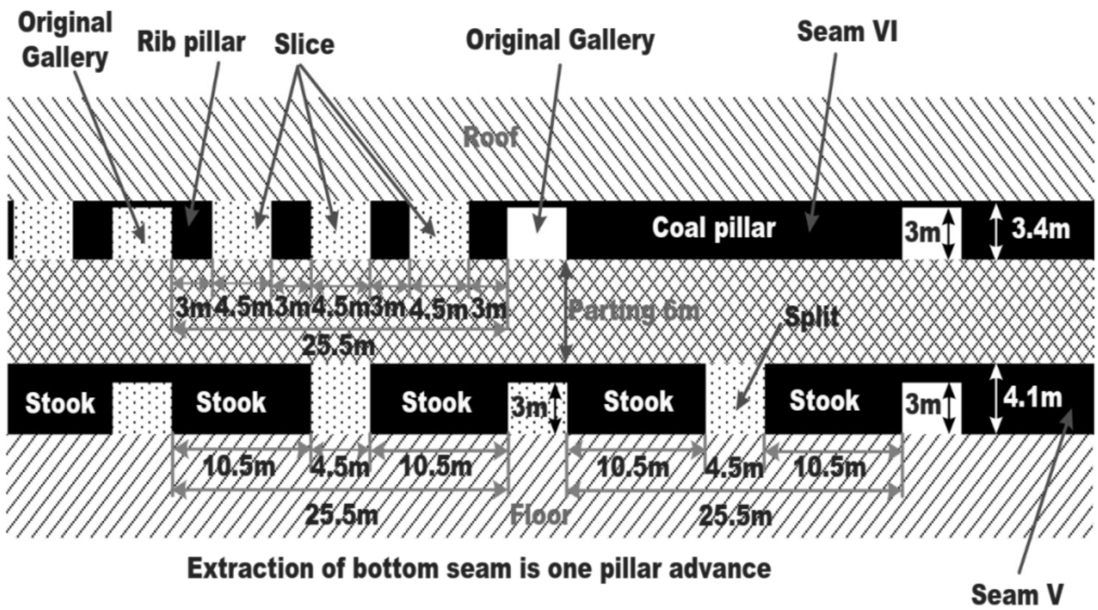


Fig. 8.3. Section illustrates the extraction of (a) top seam VI and (b) bottom seam V with hydraulic sand stowing (Das et al. 2020).

8.1.2 Special Underground Coal Mining Method with Hydraulic Stowing

The special method of underground coal mining with hydraulic stowing considering a coal seam dip of more than 30 degree is Jankowice method using bottom ash/sand or its admixture. The Jankowice method, a variant of the inclined slicing method, is a manual longwall mining technique used in India, particularly for de-pillaring (removing pillars of coal) in thick and steeply inclined coal seams (Chowdhary et al. 1975).

It involves extracting coal in multiple lifts, with a focus on hydraulic stowing to prevent roof falls. Jankowice is an inclined slicing method, where coal is extracted in slices, typically from the bottom up direction. The Jankowice method typically extracts the seam in multiple lifts, with each lift involving a series of sections or slices. The Jankowice method is used in Sudamdih mine, BCCL and Chasnalla Colliery, SAIL for both deep mines and upper seams, with hydraulic stowing. The method with stowing along with a case study have been explained hereafter.

8.1.2.1 Sudamdih mine, BCCL (Jankowice method with stowing)

Sudamdih mine, located in the Jharia coalfields of Dhanbad, Jharkhand, is an old underground coal mine operated by Bharat Coking Coal Limited (BCCL). It is a part of the Eastern Jharia Area and known for its challenging working conditions and the presence of high-quality prime coking coal in the lower seams.

The mine was overlain by many surface features like railway lines and built-up areas. The mine was operational at a depth of approximately 260m, with a thick seam of 7.3m and slice thickness of 3.5m, having gradient 35°, with immediate roof and floor being medium grained sand stone.

As shown in Fig. 8.4 the seam was developed in horizons driven through coal along the strike of the seam 100m vertically apart. The horizons were interconnected by rises of 3.5m x 3.0m driven along the floor. The strike interval between the rises was nearly 100m, forming a pair of blocks 100m x 100m plan dimensions. The lower horizon served as the main intake and coal transport road while the upper horizon served as material supply, stowing line and return airways. The central line was equipped with a chain conveyor to transport coal from two adjacent panels.

The seam was extracted in two lifts in ascending order, each 3.5m thick, one after the other. The strike face moved upwards to the rise, leaving a 20m thick barrier against the 300m horizon. Two such blocks were operated simultaneously with 25m staggering to maintain a regular flow of coal and stowing cycle. The top slice was extracted after the extraction and stowing of the bottom slices. Alternatively, the second lift extraction could be started after at least a 30m advance of the bottom slice, through the same rise.

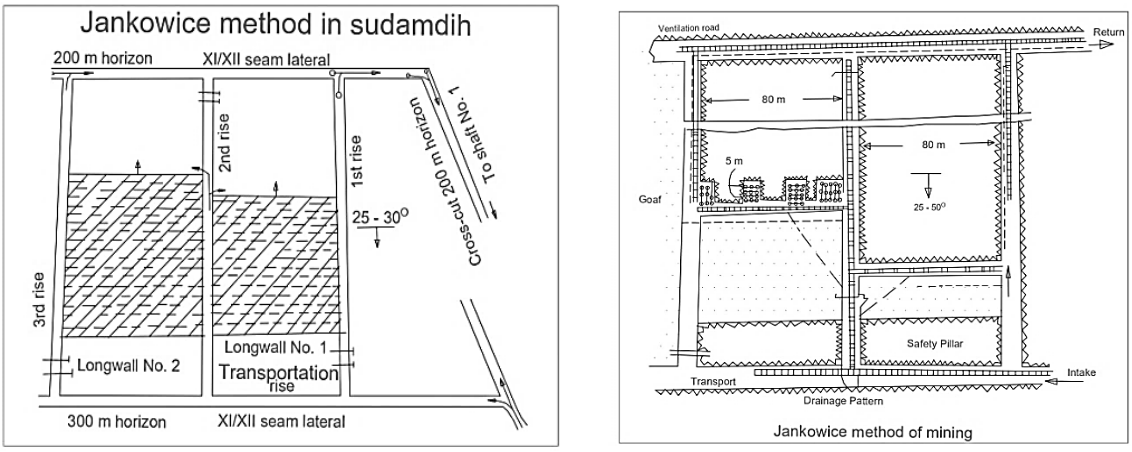


Fig. 8.4. Jankowice method of mining

8.1.2.2 Chasnalla Colliery, SAIL Deep Mine - 13/14 Seam (Jankowice method with stowing)

The extraction for Panel LT – 5 Series (between -92.5MRL and -134.9 MRL) was done by Jankowice method using bottom ash / ash or sand or its admixture. Before commencing extraction, care was taken that the face in the same seam or any other seam kept free from water before and during the extraction of the panel. The coal was extracted in three lifts and in each lift not more than true thickness of the seam (2.4m in LT-5C & 5D panel) and the parting between the lifts was 0.6 m. The faces were advanced at a uniform rate and extraction of immediate overlying lift was done after the bottom lift has advanced by 20 m ahead. The rate of face advance was so maintained that at any stage the maximum span from stowed area to the face does not exceed 5.5 m. Before commencement of extraction of coal, the panels were divided into blocks, each of maximum 30 m (MRL from -92.5 to -134.9) in length along the strike direction. Block along the strike was made by driving staples of length not more than 5.5 m measured from barricade and the width of breasting not exceeding 4.5 m. Barricades were constructed in dip side at two corners of blocks against the previous left barricade pillars and newly formed staples as shown in the Fig. 8.5.

After extracting each block, the voids created was stowed up-to 4.5 m from the stowed goaf and by leaving 1.0 m clearance between the face and barricade. Extraction in the next block was started only if adjoining voids have been stowed completely. Before initiating stowing operation, the wooden flumes were installed on the floor for proper drainage of stowed water. As the stowed mass reached the height of flumes, another layer of flumes was placed on it. Barricades of sufficient strength (hessian cloths / bamboo matting / locally available material) erected and overlapping between them was ensured to retain the stowed mass. To meet the D.G.M.S. regulation, the rate of extraction and stowing was so designed that the entire operation of one cycle could be finished within 36 hours and of one lift within 6 months.

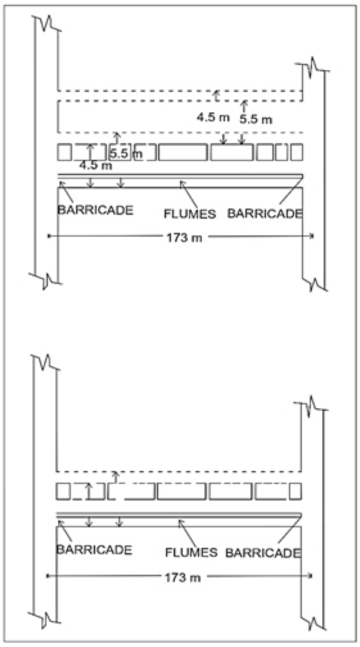


Fig. 8.5 Method with Barricade position

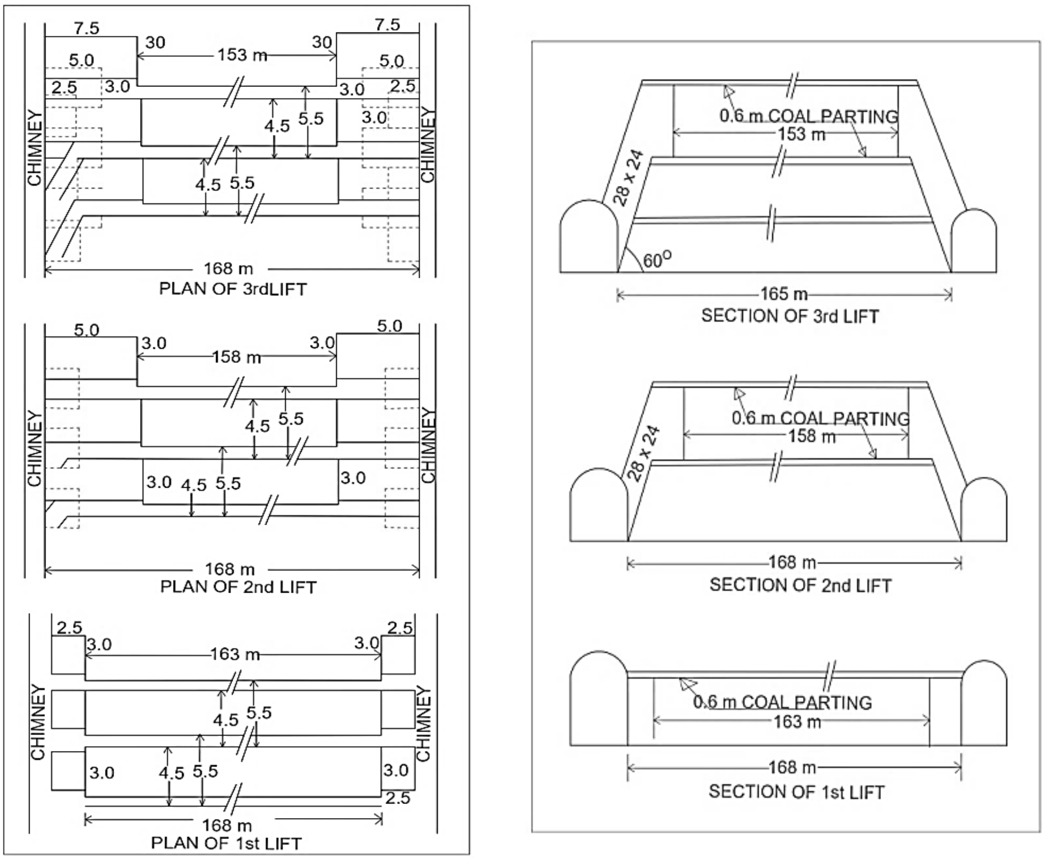


Fig. 8.6. Section showing the stage wise extraction of 1st, 2nd and 3rd lift operation of 13/14 seam

Similarly, during the extraction of 16 seam LE1 by Jankowice method, the maximum span from stowed material to the face was approximately 5.0 to 5.5 m. After extraction of 100 m along strike direction and 6.0 m along dip the void created was stowed with bottom ash (100%). Before initiating stowing operation, the wooden flumes were installed on the floor for proper drainage of stowed water. As the stowed mass reached the height of flumes, another layer of flumes was also placed on it. Barricades were erected as per the below schematic drawing (Fig. 8.7).

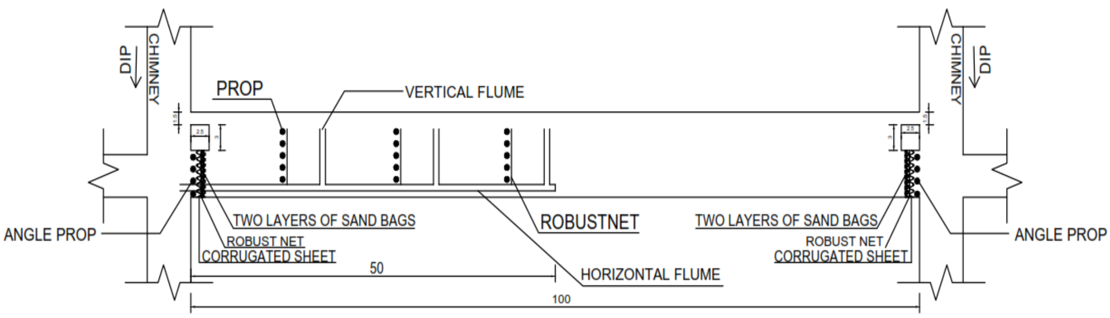


Fig. 8.7. Barricade design for Chasnalla Colliery, SAIL